Learning Objective 3.2: I can derive the array factor for an arbitrary linear array and apply pattern multiplication.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Reading a sweep.** A lab group sweeps the PHASER with all eight elements active and a source at boresight. Working at 10.3 GHz with element spacing $d = 14$ mm, they record the trace below. The sweep step is 2.8125° and the peak reads -4.1 dBFS.

θ (°)	0.0	2.8	5.6	8.4	11.3	14.1	16.9	19.7	22.5	25.3
Power (dBFS)	-4.1	-4.6	-6.2	-9.1	-14.3	-24.1	-22.0	-17.4	-16.6	-18.1

   (a) Using linear interpolation between the two samples that bracket the half-power level, find the half-power beamwidth. The trace is symmetric about 0°.

   θ_{HP} = ____

   (b) Find the first-null beamwidth as the sweep reports it, and compare it with FNBW $= 2 \arcsin(\lambda / N d)$.

   $\text{FNBW}_{\text{meas}}$ = ____

   $\text{FNBW}_{\text{calc}}$ = ____

   (c) Find the first sidelobe level relative to the peak, in dBc.

   SLL = ____

   (d) No sample in the sweep reads below about -24 dBFS, which is 20 dB below the peak, yet the noise floor on this kit is 23 dB below the peak. Explain in one or two sentences why the deepest sample sits 3 dB above the floor rather than on it.

2. **Predict before you measure.** The same array is reconfigured by setting some element gains to zero. The spacing $d = 14$ mm and the frequency 10.3 GHz do not change, so $\lambda = 29.1$ mm throughout.

   (a) Rx1, Rx2, Rx7 and Rx8 are set to zero, leaving Rx3 through Rx6 active — four adjacent elements. Predict the half-power beamwidth from $\theta_{\text{HP}} \approx 0.886\lambda/(Nd)$.

   θ_{HP} =

   (b) Now only Rx3, Rx4 and Rx5 are left on — three adjacent elements. Predict the half-power beamwidth and the first-null beamwidth.

   θ_{HP} =

   FNBW =

   (c) By how many dB should the peak of the three-element sweep sit below the peak of the full eight-element sweep, if every element gain that is still on remains at 100%?

   Δ =

   (d) A student proposes reaching the same 35° beamwidth by leaving all eight elements on and turning every gain down to 37.5%. Explain in one or two sentences why that does not work.

3. **Reconciling a sidelobe.** A second group measures the first sidelobe of the full eight-element array at -11.5 dBc, while the array factor for $N = 8$ predicts -12.8 dB. Their noise floor sits at -23 dBc and the sweep step is $2.8125°$.

   (a) The floor adds noise power to the sidelobe power the receiver reports. Treating the two as uncorrelated, compute the level the sweep should report for a true -12.8 dB sidelobe sitting on a -23 dBc floor.

   SLL = ____

   (b) Their peak reading is also affected. The peak is 23 dB above the floor by definition. By how much does the floor lift the peak reading, and which way does that push the reported sidelobe level in dBc?

   peak lift = ____

   (c) The floor explains 0.4 dB of the 1.3 dB discrepancy. Name two other contributions that plausibly make up the rest, and say for each whether it would move the reading up or down.

4. **The two-element case.** With only Rx4 and Rx5 active the array has two elements at $d = 14$ mm, and at 10.3 GHz the array factor is $AF = \cos(\psi/2)$ with $\psi = kd\sin(\theta)$.

   (a) Find the half-power beamwidth exactly, without using the $0.886\lambda/(Nd)$ rule.

   θ_{HP} =

   (b) Show that the $0.886\lambda/(Nd)$ rule fails here, and state the percentage error it makes.

   error =

   (c) Evaluate $\lambda/2d$ and use it to explain why the lab table lists the first-null beamwidth of this array as 180°.

   $\lambda/2d$ =

   (d) How far below its own peak does this two-element pattern fall at $\theta = 90^\circ$? Compare that with the 23 dB of dynamic range the sweep has above the noise floor at eight elements, and say what the trace will look like near the edges of the scan.

   drop at 90° =

5. **Design judgment.** Your section is asked to demonstrate a 20° beam on the PHASER at 10.3 GHz, using only the Element Gains controls.

   (a) How many adjacent elements must be active to give a half-power beamwidth closest to 20°? Give the aperture and the resulting beamwidth.

   N = ____

   θ_{HP} = ____

   (b) What does this cost in peak received level compared with the full eight-element array, and how much dynamic range is left above the noise floor?

   Δ = ____

   range = ____

   (c) A classmate suggests instead applying a Hann taper to all eight elements, since that also widens the beam to about 19.5°. State one advantage and one disadvantage of the taper against switching elements off, and say which you would choose for a demonstration whose purpose is to show that beamwidth depends on aperture.

Documentation: _____